

MLB WAR Predictions

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March 3, 2025

1 Abstract

Wins Above Replacement (WAR)^[1] is a comprehensive metric used to evaluate an individual baseball player’s performance and contribution to team success. In this study, we develop predictive models to estimate a Major League Baseball (MLB) pitcher’s annual WAR based on in-game performance metrics and pitching characteristics. Using compiled pitch-by-pitch and game-level information from a complete list of MLB pitchers, we first construct models to predict a pitcher’s current season WAR from observed season statistics. We then extend this analysis to forecast the following season’s WAR, providing insight into a pitcher’s projected value. Given that our dataset includes detailed pitch information, we leverage PitchFx-derived pitch classifications to assess their impact on WAR. The study aims to identify key performance indicators that contribute to a pitcher’s value and to explore the stability and predictability of WAR over time. Future applications of this research include aiding front-office decision-making in player evaluation and contract negotiations.

2 Introduction

In Major League Baseball (MLB), evaluating a pitcher’s overall contribution to team success is central to both game-day strategy and front-office decision-making. Traditional statistics—such as Earned Run Average (ERA) or strikeout totals—capture facets of performance but can overlook a pitcher’s broader impact on team outcomes. By contrast, Wins Above Replacement (WAR) condenses a player’s on-field value into a single number, comparing them against a hypothetical “replacement-level” player. As a result, WAR has emerged as a critical tool for contract negotiations, roster planning, and long-range forecasts of player performance. Accurately predicting WAR, however, remains a significant challenge. A pitcher’s season-long value reflects many interdependent factors, from pitching mechanics and defensive support to game contexts and usage patterns. Recent advancements in tracking technology—such as PitchFx—now provide pitch-by-pitch data, enabling deeper exploration of how specific pitching characteristics translate into WAR. Building on this rich data, our study develops predictive models that integrate both game-level and pitch-level metrics to estimate a pitcher’s WAR. We focus on two main objectives:

1. Current-Season Prediction: Estimating a pitcher’s WAR for the same year, based on that season’s statistics.
2. Next-Season Forecasting: Projecting WAR for the following year, using prior-season metrics.

Through experimentation with multiple modeling techniques (Random Forest, Lasso Regression, and Multiple Linear Regression), we ultimately found that a multiple linear regression model provides the most balanced combination of interpretability and predictive accuracy. Our final model explains over 93% of the variance in pitcher WAR (R-squared is 0.93) with a Root Mean Square Error (RMSE) near 0.37. While forecasting WAR one season ahead naturally introduces more uncertainty, the model still captures essential trends and distributions.

Initially by examining factors such as strikeouts, home runs allowed, total outs, and a pitcher’s role as a starter or reliever, we aim to identify key statistical drivers of WAR. These findings can help MLB teams better allocate their pitching resources, negotiate contracts, and plan rosters for future seasons. Although no metric can fully encapsulate a pitcher’s total value, our results underscore that combining traditional counting stats with granular pitch-level data yields a more holistic and actionable view of pitcher performance.

3 Data Cleaning

We first narrowed our original dataset of pitchers across all leagues by filtering primarily for the 2024 calendar year and solely players from Major League Baseball (MLB). We further refined the dataset by ensuring that only records from regular season games (Game type R) were included to ensure consistency of measurement. Additionally, we retrieved Wins Above Replacement (WAR) scores for 2024^[2] from FanGraphs, aligning them with corresponding player data. This allows for a more comprehensive understanding of each player’s contributions during the season. Some player names in the original data did not match perfectly with those in the WAR dataset, so we manually corrected these mismatches to ensure consistency and accuracy across the dataset.

Variable	Min	1st Qu.	Median	Mean	Max
total_K	0.00	7.00	26.00	41.36	220.00
total_W	0.00	2.00	8.00	12.11	67.00
total_HR	0.000	1.000	3.000	5.443	34.000
total_outs	0.0	29.0	100.0	140.9	601.0
avg_age	20.80	26.49	28.84	29.23	31.58
avg_seniority	0.00	1.000	19.00	24.26	79.00
total_games	1	6	19	24.26	79
PA	1.0	63.5	158.0	200.7	635.0
IP	0.00	10.67	36.67	50.43	208.67
Bat.WAR	-0.72998	0.00000	0.00000	0.03013	3.09396
Pit.WAR	-1.6400	-0.0598	0.1130	0.5027	6.4103
Total.WAR	-1.6400	-0.06423	0.14720	0.53226	6.41031
PlayerId	1247	15868	19918	20011	33876

Table 1: Summary Statistics for Numeric Variables

Variable	Categories
name	Abbott Andrew, Abreu Bryan, Adam Jason, Adams Austin, ...
isStarter	0, 1
Team	CIN, HOU, OAK, PHI, KCR, SDP, ...

Table 2: Categorical Variables in the Data

After cleaning, our dataset consists of 856 records (players) and 17 different columns capturing a range of pitching statistics, such as strikeouts (total_K), wins (total_W), home runs allowed (total_HR), and other relevant metrics like average age (avg_age) and average seniority (avg_seniority). We also included a column indicating whether or not the player is a starter (isStarter) and their number of games played (total_games). The integration of Batting WAR (Bat.WAR), Pitching WAR (Pit.WAR), and Total WAR provides a more holistic view of each player’s performance in the 2024 season.

Variable	Sale Chris	Skubal Tarik	Wheeler Zack	Ragans Cole	Cease Dylan
total_K	212	220	211	217	215
total_W	30	20	39	52	50
total_HR	8	13	18	10	16
total_outs	503	550	574	527	535
avg_age	35.36499	27.69370	34.19660	26.62420	28.59646
avg_seniority	13.939276	3.869638	11.064732	1.896291	5.003164
isStarter	1	1	1	1	1
total_games	29	31	32	32	33
Name	Chris Sale	Tarik Skubal	Zack Wheeler	Cole Ragans	Dylan Cease
Team	ATL	DET	PHI	KCR	SDP
PA	NA	NA	NA	NA	NA
IP	177.6666	192.0000	200.0000	186.3333	189.3333
Bat.WAR	0	0	0	0	0
Pit.WAR	6.410311	5.860317	5.428380	4.880327	4.761360
Total.WAR	6.410311	5.860317	5.428380	4.880327	4.761360
PlayerId	10603	22267	10310	21846	18525

Table 3: Top 5 Pitchers by Total WAR (Horizontal Layout)

This table shows the Top 5 pitchers by Total WAR, showing the new data structure. It highlights the top 5 pitchers by Total WAR, all of whom are starters, reinforcing the idea that starting pitchers tend to accumulate higher WAR. While all five pitchers have strong strikeout numbers, their wins and home runs allowed vary, suggesting different pitching styles. Notably, Chris Sale, the most senior pitcher, leads in WAR despite pitching fewer innings than some others, indicating high efficiency. The data also show that both experienced and younger pitchers can achieve high WAR. Further analysis could examine which performance metrics contribute most to WAR.

4 Exploratory Data Analysis

Now, we can conduct some exploratory analysis to better understand the key factors affecting pitcher's WAR. Primarily we will focus on the relationship between the new WAR (Wins Above Replacement) metric and various predictors such as starter status, strikeouts, home runs allowed and other relevant pitching statistics. The goal of this analysis is to identify key correlations, trends and patterns to help understand the drivers of WAR, which is an important measure of a player's overall contribution to the team. With this EDA, we aim to gain valuable insights for further analysis or model development.

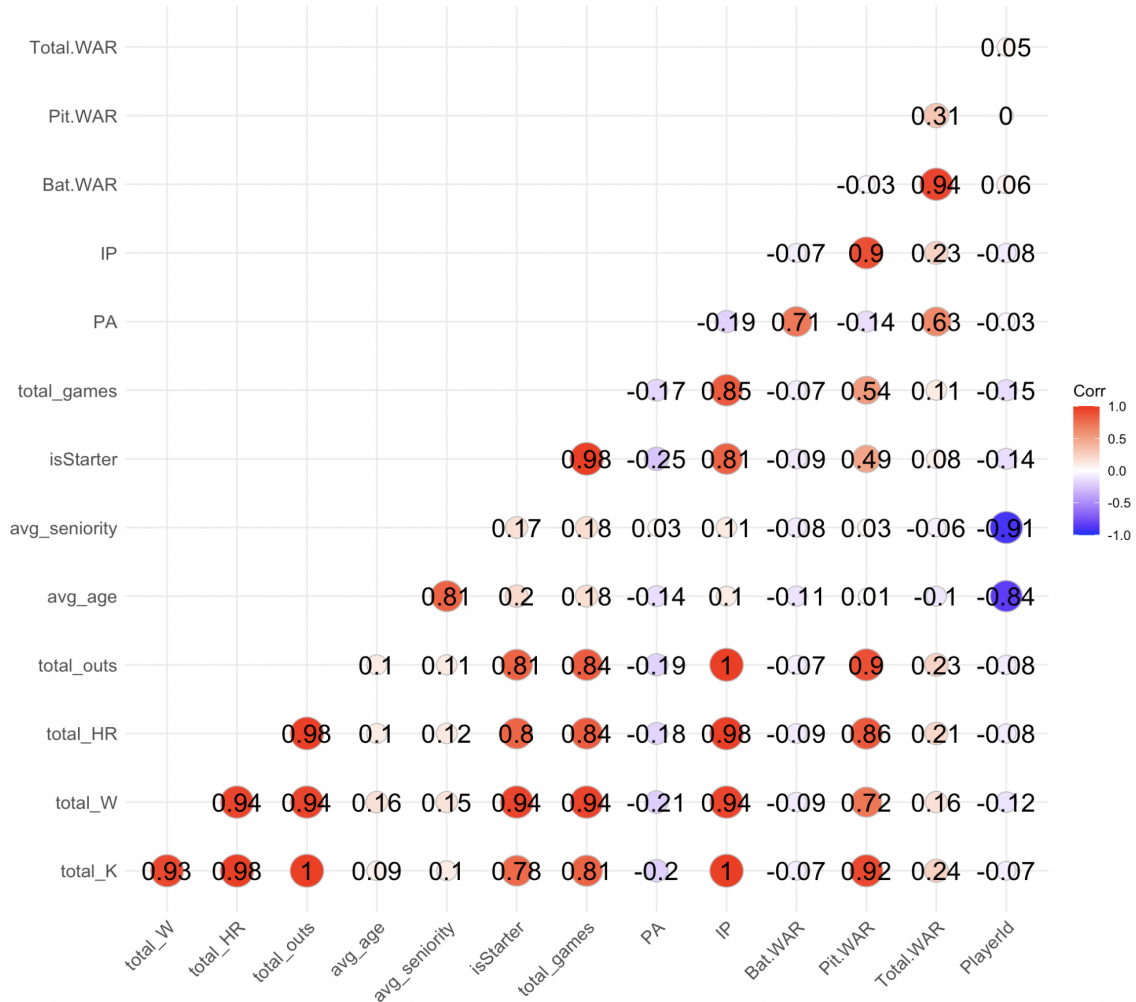


Figure 1: Correlation Matrix of Pitching Metrics

This correlation matrix confirms that strikeouts (total_K) and wins (total_W) have a strong positive correlation with Pitching WAR (Pit.WAR) (0.77-0.78), making them key predictors. Home runs allowed (total_HR) also has a high correlation with strikeouts and wins, indicating potential multicollinearity issues. Additionally, age and seniority show a negative correlation with WAR, suggesting that older and more experienced pitchers tend to have lower WAR values, possibly due to declining performance.

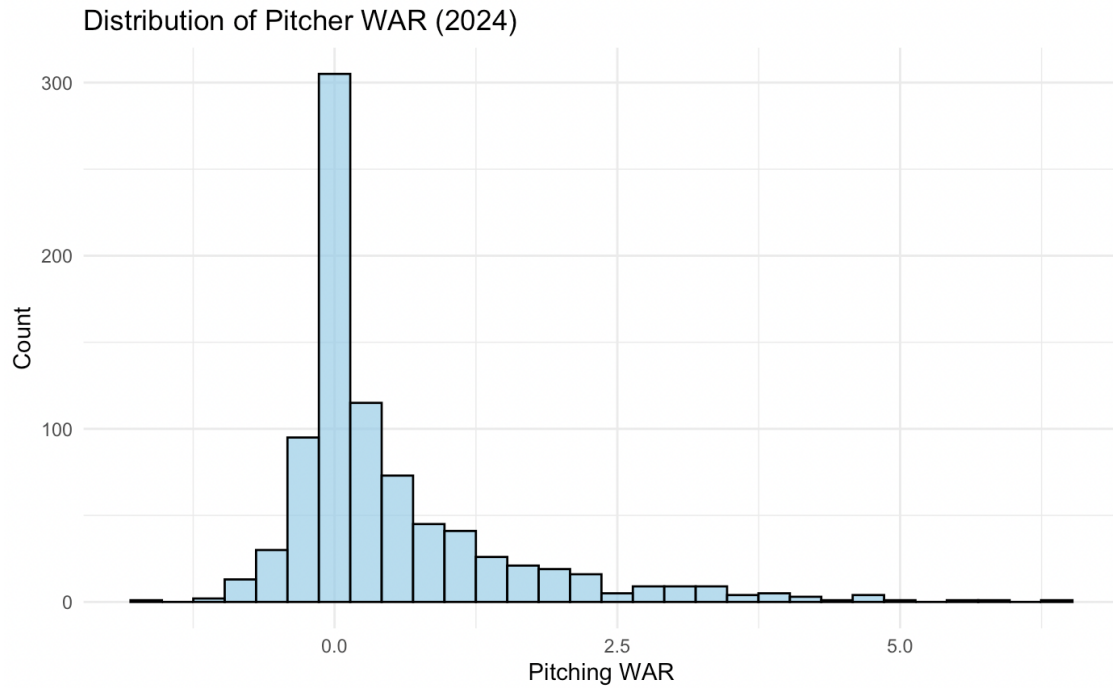


Figure 2: Pitching WAR Bar Plot

This plot summarizes the distribution of Pitcher WAR during the 2024 calendar year. We can see that most pitchers lie in the lower WAR values around zero, while very few pitchers have achieved impressive WAR statistics as we increase in the x-direction (WAR).

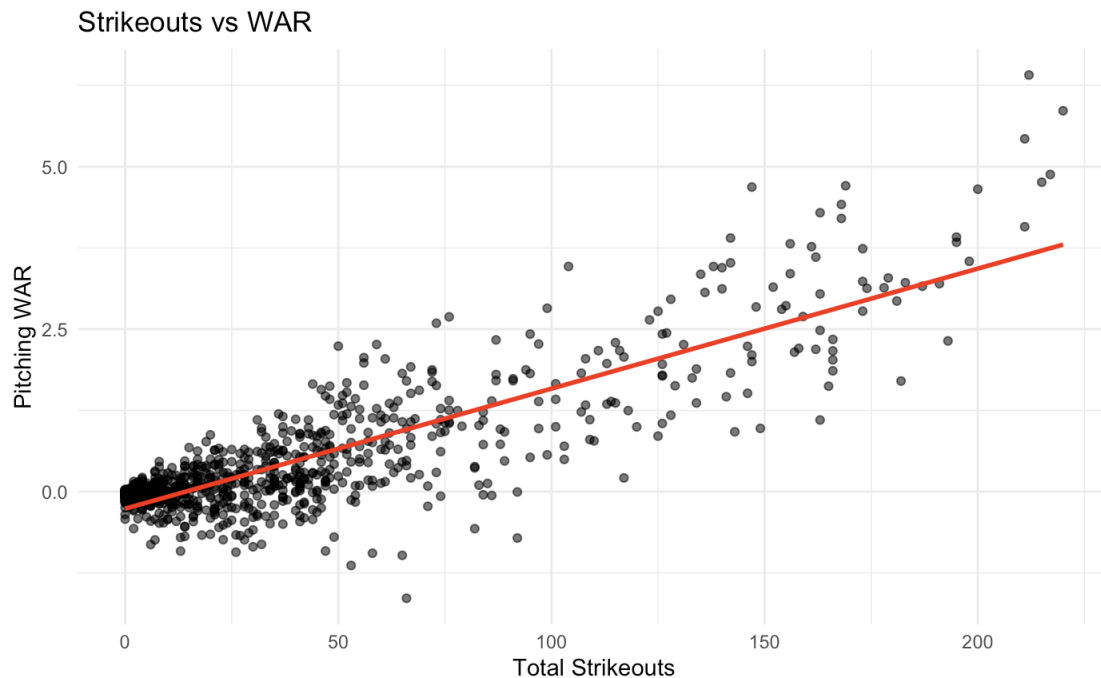


Figure 3: Scatter Plot for Strikeouts vs. WAR

The scatter plot shows the relationship between total Strikeouts and Pitching WAR. The line suggests a positive correlation between these two variables, meaning that, as the number of strikes increases, the Pitching WAR will increase as well. This implies that pitchers with higher strikeouts tend to have better overall performance (as measured by WAR).

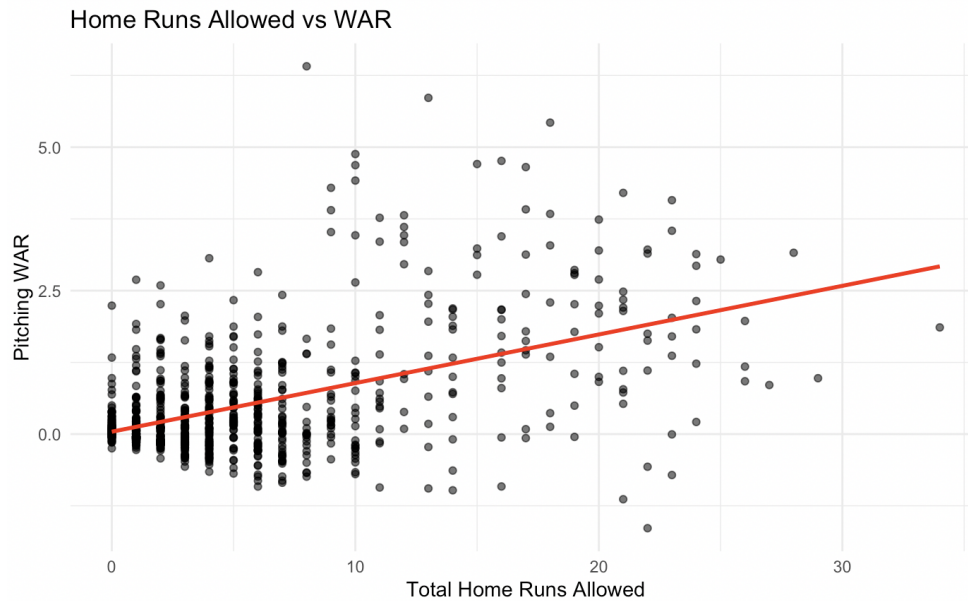


Figure 4: Scatter Plot for Total Home Runs Allowed vs. WAR

The scatter plot shows the relationship between Total Home Runs Allowed and Pitching WAR. The line suggests a positive correlation, meaning that as the number of home runs allowed increases, the Pitching WAR tends to increase as well, although there is some spread in the data, indicating that other factors are influencing the relationship.

Also, the results are quite similar for the total number of outs a pitcher has recorded and the total number of wins. Both of these variables also show a positive correlation with Pitching WAR, suggesting that as these metrics increase, the Pitching WAR tends to rise as well. This indicates a general trend where better performance in these key areas often correlates with a higher WAR for pitchers.

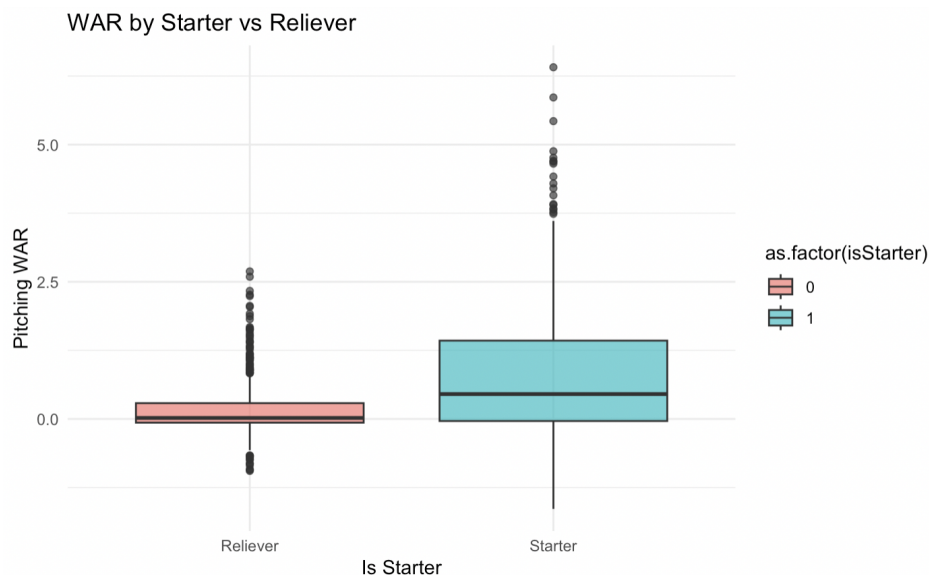


Figure 5: Boxplot for Starters vs. WAR

The boxplot shows that starting pitchers generally have higher WAR than relievers, with a wider range of values. The median WAR for starters is higher, and they also have more extreme high-value outliers, indicating that top-performing pitchers are usually starters. In contrast, relievers have a more concentrated distribution with fewer high-WAR players. This suggests that starters contribute more significantly to team performance. Further statistical tests could confirm whether this difference is significant.

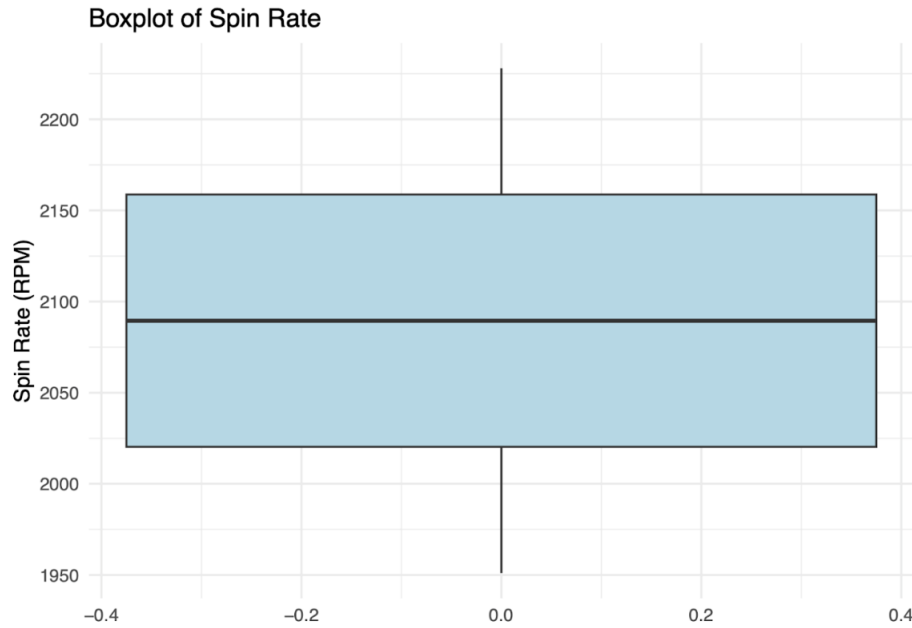


Figure 6: Boxplot of Spin Rate

In the boxplot, the thick black line represents the median spin rate, which is about 2100 RPM (Spin Rate). The box shows the interquartile range, spanning from a little above 2000 to 2150 RPM. The whiskers extend slightly beyond the interquartile range, but extreme outliers are not visible. In comparison to the histogram of spin rate, the range appears to be more normally distributed than the histogram.

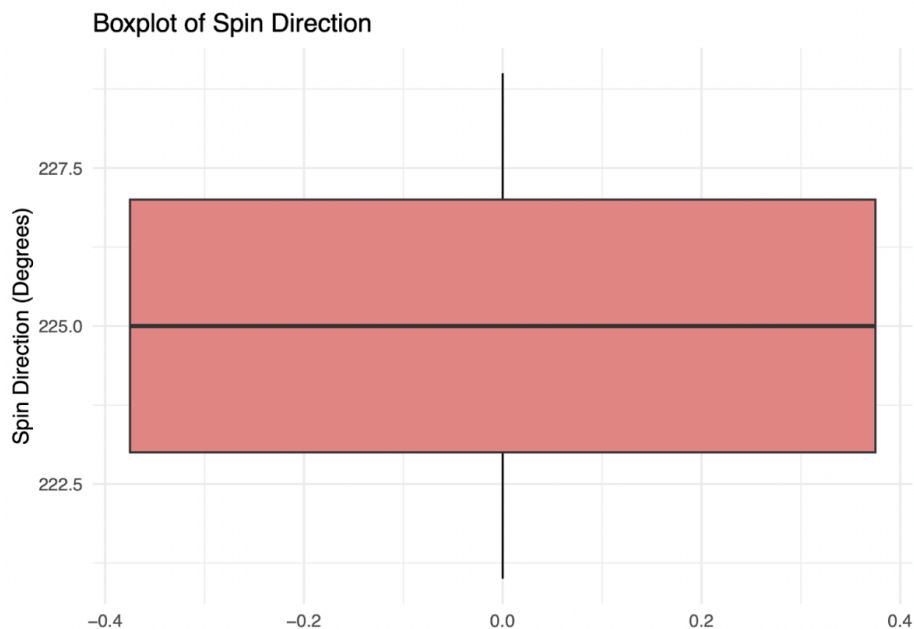


Figure 7: Boxplot of Spin Direction

The median spin direction is around 225 degrees, which tells us that most values are clustered around this range. The interquartile range shows values between 222.5 and 227.5 degrees. The spread is relatively balanced, but the interquartile range does show extreme outliers. This suggests that the range of spin direction values is small and consistent.

5 Model Selection

5.1 Multi-linear regression

We began with a multiple linear regression, in which we included all available predictors to see how they related to Pit.WAR. After evaluating the model using a significance plot, we found that only the predictors total_K, total_W, total_HR, total_outs, isStarter and total_games were statistically significant, as evidenced by the fact that all of their p-values were less than 0.05.

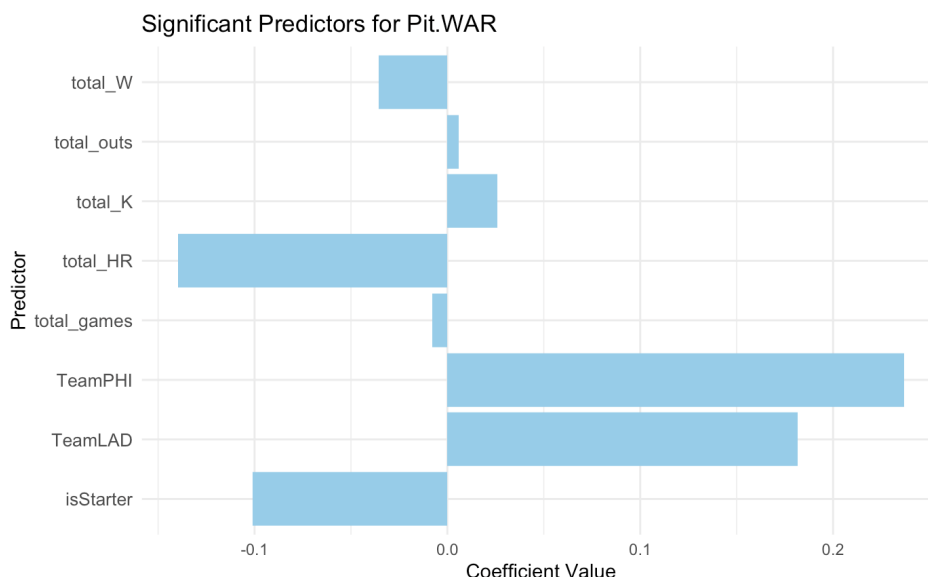


Figure 8: Significant Predictors by Multi-Linear Regression Model

With this in mind, we decided to simplify the model by focusing only on these significant predictors to ensure that the model is more efficient and easier to interpret while maintaining predictive accuracy.

Variable	Estimate	Std. Error	t value	Pr(> t)	Sign
Intercept	-0.0397146	0.0164444	-2.415	0.0159	*
total_K	0.0262969	0.0007608	34.566	$< 2e - 16$	***
total_W	-0.0360197	0.0014482	-24.872	$< 2e - 16$	***
total_HR	-0.1368877	0.0033678	-40.646	$< 2e - 16$	***
total_outs	0.0056960	0.0003180	17.909	$< 2e - 16$	***
isStarter	-0.0441718	0.0237403	-1.861	0.0631	
total_games	-0.0060745	0.0005225	-11.626	$< 2e - 16$	***

Table 4: Linear Regression Coefficients for Pit.WAR Prediction

Metric	Value
Residual Standard Error	0.261 on 848 degrees of freedom
Multiple R-squared	0.9348
Adjusted R-squared	0.9344
F-statistic	2027 on 6 and 848 DF
p-value	$< 2.2e - 16$
RMSE	0.3726398

Table 5: Linear Regression Model Summary

This multi-linear regression model shows strong predictive performance with an R-squared value of 0.9348. This means that the predictors in the model explain about 93.48% of the variance in the target

variable (Pit.WAR). Moreover, the RMSE (root mean square error) is 0.3726, indicating that the mean difference between predicted and true values is minimal, highlighting the a relatively high accuracy of the model. Overall, the model provided reliable predictions and showed excellent explanatory power.

5.2 Random Forest

We also used a Random Forest Regressor on the same data. The feature importance of this regressor is as shown below:

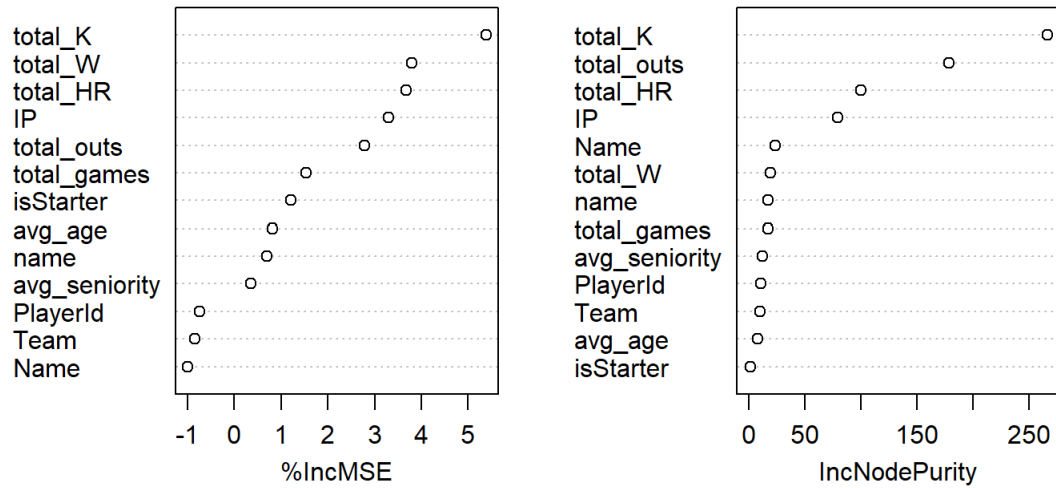


Figure 9: Significant Predictors by Feature Importance in Random Forest

The Random Forest Regression model shows okay predictive performance with an Root Mean Squared Error value of 1.38. The R-squared of this model is 0.97. The model's prediction error is quite high, showing that it does not have high accuracy and this model does not fit the data very well, and will not be used for forecasting. The reason for the high R-squared value and high RMSE is likely due to the model overfitting.

5.3 Lasso Regression Model

We performed a Lasso regression using the same dataset to identify the most important predictors for the model, because it regularizes the model by penalizing the coefficients of less significant features, allowing us to focus on the most relevant variables.

Variable	Coefficient
Intercept	0.028721531
total_K	0.026912490
total_W	-0.034185644
total_HR	-0.131563456
totalOuts	0.005066239
avg_age	.
avg_seniority	0.005469390
isStarter	-0.087971140
total_games	-0.006941521
Team	.
IP	.
RMSE (Lasso model)	0.3566308

Table 6: Lasso Model Coefficients

The Lasso regression results show the coefficients of the different predictors in the model, with some variables contributing positively and others negatively. The model provides insights into which factors

are most influential in predicting the target variable. The RMSE of the Lasso model is 0.3566308, indicating a reasonable fit to the data, though there is still room for improvement in model accuracy.

5.4 Comparison

After running the three models, Multi linear regression, Random Forest and Lasso regression, we decided to use R-squared and RMSE as evaluation measures for comparing the models. R-squared gives us insight into the proportion of variance explained by the model, thus evaluating the explanatory power of the model. RMSE, on the other hand, helps us understand the average error between the predicted and actual values, giving us a clear indication of the model's accuracy. By analyzing these two metrics, we can make a more informed decision about which model performs best.

Model	R-squared	RMSE
Linear Regression	0.9348310	0.3726398
Random Forest	0.9753818	1.3872194
Lasso Regression	0.9276461	0.3566308

Table 7: Comparison of Models: R-squared and RMSE

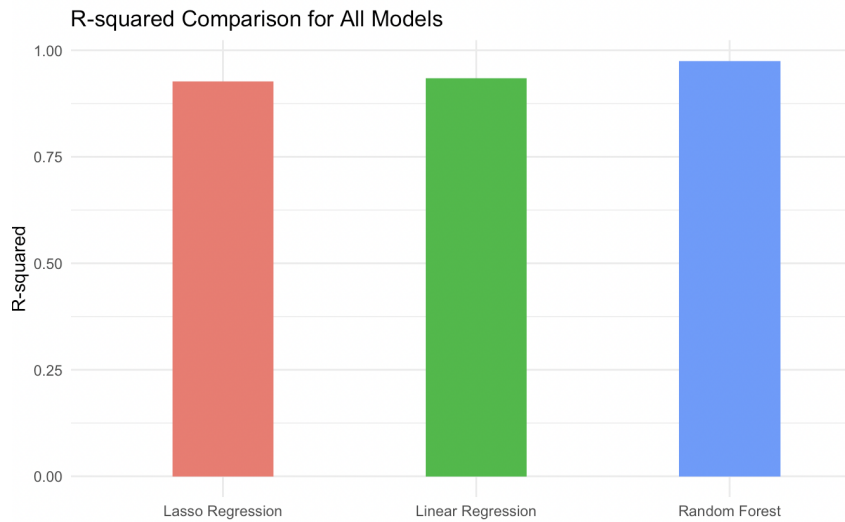


Figure 10: R-squared Comparison of Models

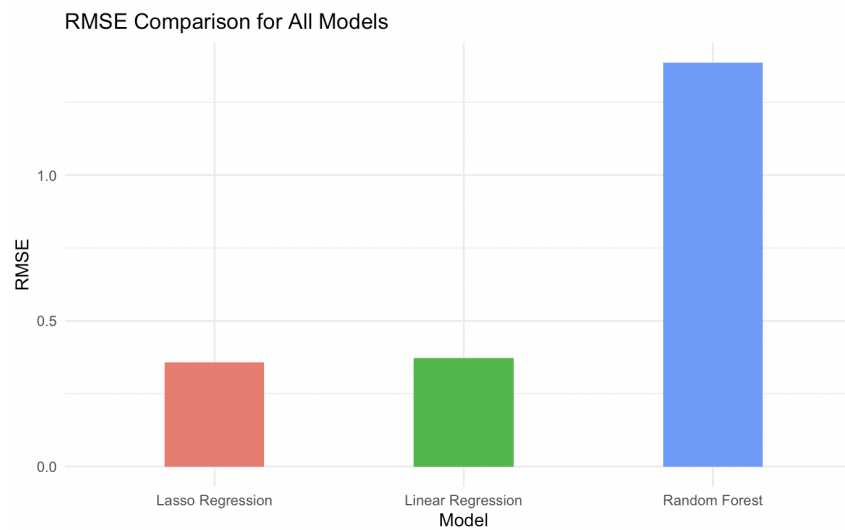


Figure 11: RMSE Comparison of Models

Based on the results of both R-squared and RMSE, the Random Forest model has the highest R-squared value of 0.9753818, indicating that it explains the largest amount of variance in the data. However, the RMSE value of 1.3872194 is quite high, indicating that the model has a higher prediction error than the other models. On the other hand, the linear regression model has a slightly lower R-squared value of 0.9348310 but a much lower RMSE value of 0.3726398, indicating that although it may not explain as much variance as the Random Forest model, its predictions are much more accurate. The lasso regression model is also a strong model with an R-squared of 0.9276461 and an RMSE of 0.3566308.

Given these findings, we should choose the linear regression model because it offers a better balance between model performance and predictive accuracy. Additionally, Multilinear regression is simpler and more interpretable than Random Forest and Lasso, making it more suitable for real-world applications. Although the R-squared is slightly lower, the significantly lower RMSE suggests that it provides more reliable predictions.

6 Results

In this study, we have chosen to use a multiple linear regression model to predict the Pitcher Wins Above Replacement (WAR) based on the 2024 data. This approach allows us to examine the relationship between various performance metrics and the overall contribution of a pitcher. Equation (1) represents the model used for this prediction, where the Pitcher WAR is explained by a combination of factors such as strikeouts, wins, home runs, outs, games played, and whether the pitcher is a starter. By using this multi-linear equation, we aim to capture the key variables that influence the performance of pitchers in a comprehensive manner.

$$\begin{aligned} \text{Pit.WAR} = & -0.0397146 + 0.0262969 \cdot \text{total_K} \\ & - 0.0360197 \cdot \text{total_W} - 0.1368877 \cdot \text{total_HR} \\ & + 0.0056960 \cdot \text{total_outs} + 0.0441718 \cdot \text{total_games} \\ & - 0.0060745 \cdot \text{isStarter} \end{aligned} \quad (1)$$

Next, we will use the data from the 2023 and apply the same multi-linear equation to predict the Pitcher WAR for the 2024 season. Specifically, we will focus on players who participated in both the 2023 and 2024 seasons. By using 2023 performance metrics such as strikeouts, wins, home runs, strikeouts, games played and whether or not the player was a starter, we want to test the accuracy of this approach to predict the expected contribution of the 2024 season.

The RMSE of the forecasting model is 1.022354, which indicates that, on average, the model's predictions deviate from the actual Pitcher WAR values by approximately 1.02 units. While this level of error is relatively modest, it also reflects the inherent uncertainty and variability present in forecasting based on past data.

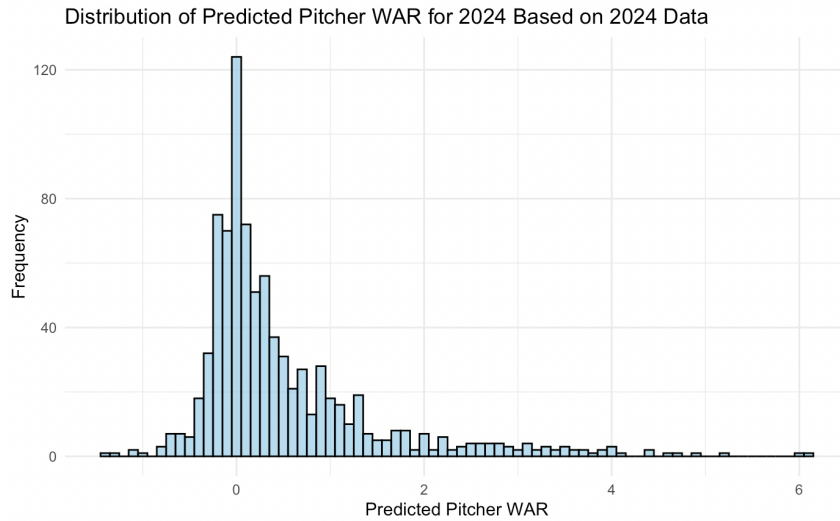


Figure 12: Predicting Pitcher WAR from 2024 Data

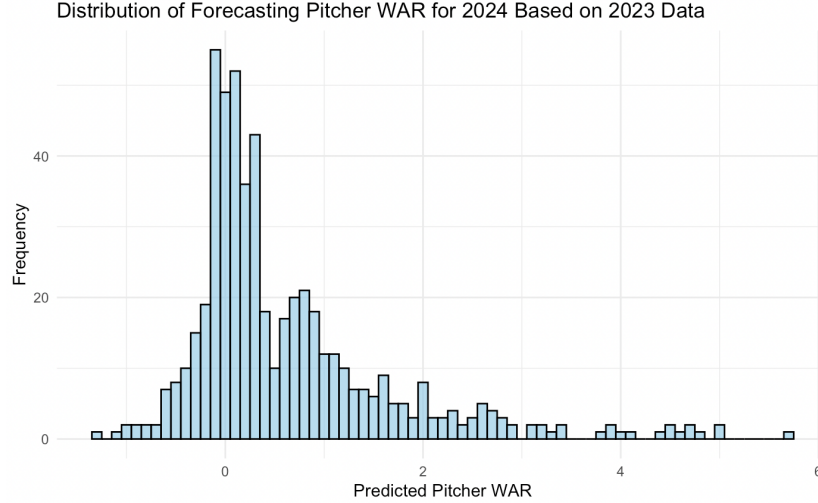


Figure 13: Forecasting Pitcher WAR from 2023 Data

In addition, we compared the predicted and forecasted WAR distribution of pitchers for 2024. These two plots show the following: (1) the prediction model using 2024 data closely mirrors the actual Pitcher WAR distribution, and (2) the prediction error for the forecasting model using 2023 data is more significant. However, despite the higher RMSE of the forecasting model, the projected distributions still follow a similar trend to the actual Pitcher WAR distribution. This indicates that the model is still effective in capturing some significant patterns in the data even though the prediction error is large.

7 Conclusion & Limitations

7.1 Conclusion

This study was aimed to develop predictive models to forecast Wins Above Replacement for pitchers by analyzing their game by game performance and pitching statistics. We used three different models, including Multilinear regression, Random Forest regression, and Lasso regression. Multilinear regression performed the best, explaining 93% of the variability in WAR.

We chose Multilinear Regression to forecast Pitcher WAR for the 2024 season from 2023 data, and specifically focused on players that played in both seasons. We identified key predictors of pitcher WAR such as strikeouts, home runs allowed, total outs, and pitching role (starter vs. reliever). The Root Mean Squared Error of the forecasting model is 1.022, however, the distribution of the forecasting model is very similar to the actual Pitcher WAR distribution. The model still captures the significant patterns of the data, despite a large prediction error.

Although our model demonstrated moderately accurate representation, it is important to consider some limitations that exist such as data availability constraints and model selection constraints. It is vital to address these limitations in future studies, as this would enhance predictive reliability through machine learning techniques and other contextual variables.

We hope our study results can contribute to front offices in player evaluation, contract negotiations, and roster planning. Future research can expand on our findings as more MLB seasons play out for pitcher data to be collected, and through incorporating more advanced modeling techniques to further improve the accuracy and reliability of WAR forecasting.

7.2 Limitations

Data Quality

Our data relies on recorded pitch-by-pitch and game-level statistics which may contain missing or inconsistent values. Possibly important factors such as weather conditions or injuries were not considered due to unavailability.

Model Selection Limitation

We only ran three different models, and selected the best-performing model from the three. However, these three models may not be the best fit for the data, and there may be a model that is better that we did not test. Ideally, we would have the resources to test a wider range of models to identify the most optimal one for accurately predicting WAR.

Potential Overfitting to Historical Trends

If pitching or evaluation criteria changes across the league, adjustments would likely need to be made to the model. In other words, the current model performs well on the available dataset, but its continued effectiveness depends on the stability and consistency of past trends.

References

- [1] *Wins above Replacement (WAR)* — *Glossary*. MLB.com, www.mlb.com/glossary/advanced-stats/wins-above-replacement.
- [2] Major League Baseball, and Sports Info Solutions. *MLB WAR Leaderboards 2024* — *FanGraphs Baseball*., FanGraphs Baseball, 2024, www.fangraphs.com/leaders/war?season=2024.